



DPUTTSServer

Electronic speech synthesis („text-to-speech“) using the operating system’s speech synthesis engine.

1. Static parameters:

<i>Engine</i>	Selects which engine is used to synthesize speech. default: (<i>empty</i>) which selects the system’s default engine (SAPI on Windows, speech-dispatcher on Linux). Usually, no other engines are available.
<i>Locale</i>	Selects the language and country of the speaker to be used. default: (<i>empty</i>) which selects the logged-in user’s language and country. The format is <i>lang_COUNTRY</i> (e.g. <i>de_DE</i> for German).
<i>Voice</i>	Selects the voice to use. default: (<i>empty</i>) selects the first available voice for the given language and country.
<i>Rate</i>	Modifies the rate of speech. default: 0.0 (normal rate of speech) Values from -1.0 (very slow) to 1.0 (very fast) are possible.
<i>Pitch</i>	Modifies the pitch of the voice. default: 0.0 (normal pitch for the selected voice) Values from -1.0 (very low) to 1.0 (very high) are possible.
<i>Volume</i>	Modifies the volume of sound output. default: 0.7 Values from 0.0 (no sound) to 1.0 (maximum volume) are possible.
<i>Port</i>	TCP port used for communication between the different speech synthesis DPUs. default: 0 which means the port given in the SILAB network configuration (<i>sys.par</i>) is used.

2. Outputs:

<i>IsSpeaking</i>	Indicates whether the DPU is currently emitting some voice output. default: false
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3. Installation:

Please note that you may need to install additional speech output software for your operating system, in order to be able to synthesize text in the selected language.

On Windows 10, use the *Settings > Time & Language > Region & Language* menu to install text-to-speech voices.



On Linux, install speech-dispatcher and an engine supported by it (e.g. *espeak* or *festival*) and configure it accordingly.

In order to see the supported values for the *Engine*, *Locale* and *Voice* parameters, you can enter "x" for each of them, which will output a list of possible values in SILAB's message window.

4. Functional principle:

DPUTTSServer opens a TCP port and waits for incoming connections.

As soon as a connection has been established, all data is read line by line, interpreted as UTF8 coded strings then synthesized using the operating system's speech synthesis engine and played back on the default audio device.

DPUTTSClient can establish such TCP connections to **DPUTTSServer** and allows to send texts given to the DPU input which are then synthesized accordingly.

DPUTTSSCNX may be loaded additionally to **DPUTTSClient** and allows to trigger playback via events of the SCN database. (See its documentation for details.)

```
DPUTTSServer TTS
```

```
{  
    Computer = {TTS};  
    Index = 20;  
    TreeviewGroup = "Sound";  
};
```

```
DPUTTSClient TTSClient
```

```
{  
    Computer = {TTS};  
    Index = 50;  
    TreeviewGroup = "Sound";  
};
```

```
DPUTTSSCNX TTSSCNX
```

```
{  
    Computer = {TTS};  
    Index = 51;  
    TreeviewGroup = "Sound";  
};
```